

Advancing Clean Energy Conversion through Electrocatalysis: High-Efficiency Oxygen Reduction in Sustainable Fuel Cells

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ABSTRACT

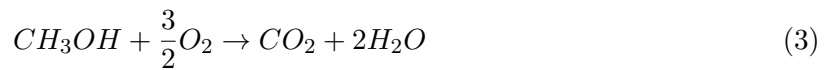
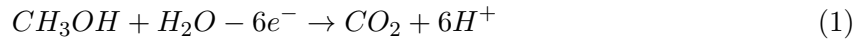
Direct methanol fuel cells (DMFCs) are a kind of fuel cells with high energy density, small size, and no pollution, and thus DMFCs have received extensive attention from countries all over the world in recent years. The reaction at the cathode of DMFCs is oxygen reduction reaction (ORR) and the kinetics of ORR is extremely slow. Accordingly, an efficient electrocatalyst is required. The currently widely used electrode is commercial Pt/C electrode. Commercial Pt/C electrode has some shortcomings such as catalyst aggregation and detachment from carbon support. Moreover, the high cost of Pt as a rare precious metal also affects its practical use in DMFCs. In this work, carbon nanotubes were used as the substrate, and after modified by melamine, CoFeMn-LDH nanosheets were formed on carbon nanotubes. The carbon nanotubes supported CoFeMn-LDH were then calcined at high temperature under the protection of nitrogen to prepare manganese-doped ferroferric oxide/cobalt catalysts loaded on nitrogen-doped carbon nanotubes (MFC@N-CNT). The characterization results show that the catalyst nanoparticles are uniformly distributed on carbon nanotubes. Meanwhile, doping of nitrogen on carbon nanotubes which have excellent conductivity and high specific surface area can effectively prevent catalyst aggregation and detachment from carbon nanotubes, and thereby improve the ORR performance of MFC@N-CNT. Although the half-wave potential of oxygen reduction of MFC@N-CNT is slightly inferior to that of the commercial Pt/C material, MFC@N-CNT has a better stability and a relatively price advantage. This study provides a new idea for the design and selection of cathode materials for DMFCs.

Keywords: Carbon nanotube, Layered double hydroxide, Non-precious metal catalyst, Oxygen reduction reaction.

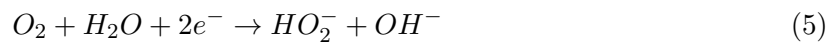
1 Introduction

With the growth of the global population and the process of industrialization among countries, the demand for clean and low-cost electrical energy resources is rising steadily[1]. However, in most countries, the dominant source of electrical energy is still generated from the combustion of fossil fuels. In recent years, an increasing number of countries have recognized the imminent scarcity of conventional fossil fuels, whose resources require thousands of years to regenerate once exhausted. In addition, the overuse of fossil fuels is leading to multiple environmental threats[2]. The combustion of coal, petroleum, and natural gas not only generates greenhouse gases, but also produces a variety of polluting gases, including sulfur dioxide and nitrogen oxides[3]. To overcome these issues, researchers are motivated to develop alternative energy resources and novel energy storage technologies. Among these potential alternatives, fuel cells have emerged as one of the most promising energy storage devices, owing to their excellent ability to convert the chemical energy of fuel compounds into electrical energy[4]. In addition, fuel cells exhibit several advantages, such as low dependency on charging processes, convenient refueling operations, and higher energy density compared with traditional fossil fuels[5]. Among various types of fuel cells, direct methanol fuel cells (DMFCs) are considered to be next-generation energy storage devices due to their compact size, convenience of storage, low cost, wide availability of liquid methanol, as well as environmentally friendly reaction products[6][7].

Among various DMFC configurations, cells composed of a carbon anode loaded with a Pt or Pt-Ru alloys catalysts, a proton exchange membrane, and a carbon cathode loaded with a Pt catalyst have been widely studied[8][9]. In such systems, the methanol oxidation reaction (MOR) occurs at the anode (Eq. 1)[9], while energy is released at the cathode through the oxygen reduction reaction (ORR) (Eq. 2)[10]. Protons generated at the anode migrate through the proton exchange membrane to the cathode, while electrons flow through the external circuit, enabling the full cell to generate electric power. The overall discharge reaction of DMFCs is summarized in Eq. 3.



Due to the intrinsically slow anodic and cathodic reaction kinetics, highly active electrocatalytic materials are required to facilitate both MOR and ORR. In alkaline electrolyte systems, ORR can proceed via two main pathways. The four-electrons pathway involves the direct reduction of O_2 to OH^- (Eq. 4)[11]. Alternatively, the two-electrons pathway reduces O_2 to HO_2^- (Eq. 5), followed by the subsequent conversion of the intermediate to OH^- (Eq. 6)[12]. In practical DMFCs, both pathways occur simultaneously.



Considering the comparatively slower kinetics of ORR, highly active electrocatalytic materials are crucial to facilitate this process. Pt metal and Pt-Ru alloys have been widely studied as high-performance ORR catalysts, and commercial Pt/C electrodes have been employed in

practical DMFCs[13]. However, the extremely high cost of precious metal catalysts limits the widespread commercialization of DMFCs. To address this challenge, in recent years, studies focusing on low-cost ORR catalysts, including oxides, sulfides, and carbides of transition metals (e.g., Fe, Co, Mn, Ni) as well as carbon-based materials, have been conducted[14]. In addition, composite electrode materials consisting of carbon-based materials loaded with transition metal oxides or hydroxides have been explored, showing promising ORR electrocatalytic performance and offering cost-effective alternatives to commercial Pt/C cathode[15].

Nanomaterials, referring to materials with at least one dimension on the nanometer scale, exhibit distinct physical and chemical properties compared to their bulk counterparts, such as enhanced mechanical, electrochemical, optical, and magnetic properties. Carbon nanotubes (CNTs), with diameters of tens of nanometers and lengths of a few micrometers, are considered as typical one-dimensional nanomaterials. Compared with other carbon-based materials, CNTs exhibit several advantages such as ease of synthesis, high specific surface area, satisfactory electron transport efficiency, and the ability to withstand rapid volumetric expansion and contraction during chemical reactions[16]. Although CNTs cannot directly catalyze the ORR due to their low catalytic activity, their uniform tubular structures and large surface areas allow CNT frameworks to serve as supports for electrocatalytic nanoparticles, indicating their potential as carbon-based substrates for ORR composite catalysts[17]. In addition, it has been reported that nitrogen doping in CNTs can enhance electron mobility and electrochemical performance, while the lattice structure of CNTs is not destroyed during the N-doping process[18].

Metallic catalysts at nanoscale are commonly synthesized by calcining metal hydroxides at high temperatures. To prepare nanomaterials consisting of multiple metal oxides or elements, layered double hydroxides (LDHs) can be used as precursors. LDHs are layered hydroxides composed of two or more metal cations, anions, and water molecules. The positively charged layers are balanced by interlayer anions, rendering the overall structure electrically neutral. The composition and ratio of metal cations in LDHs can be flexibly adjusted, as well as the exchangeable anions, making LDHs suitable for applications in chemical adsorption and electrocatalysis[19]. As two-dimensional nanomaterials, the size and morphology of LDHs are highly sensitive to synthesis conditions, such as pH, reaction temperature, preparation method, and the choice of metal cations. Therefore, optimizing synthesis conditions is crucial to obtain LDHs with nanoscale dimensions, well-defined morphology, and superior electrocatalytic performance[20].

To suppress the aggregation of nanoscale catalysts, previous research in our group has shown that introducing CNTs as nucleation substrates and growth sites for LDHs can effectively reduce aggregation, thereby enhancing electrocatalytic performance compared with bulk catalysts[21]. Therefore, to synthesize a CNT-based ORR catalyst with transition metal oxide loading, it is worthwhile to attempt growing LDHs on the surface of CNTs followed by high-temperature calcination of the composites. In addition, when melamine-derived LDHs are calcined at an appropriate temperature, it is possible to dope nitrogen from melamine into CNTs, generating N-doped CNTs and further improving the electrochemical performance of the composite materials.

In this study, a novel Mn-doped $\text{Fe}_3\text{O}_4/\text{Co}$ loading of N-doped CNT composite material (MFC@N-CNT) is synthesized and characterized. X-ray diffraction (XRD), scanning electron microscopy (SEM), and scanning transmission electron microscopy (STEM) are used to characterize the chemical composition and morphology of the material. Electrochemical measurements are conducted using MFC@N-CNT and commercial Pt/C electrodes. The results show that MFC@N-CNT exhibits comparable electrocatalytic ORR performance to commercial Pt/C electrodes, while demonstrating excellent cost-effectiveness and cycling stability. The electrochemical performances of composite catalysts synthesized under different conditions are also compared with those of MFC@N-CNT, highlighting the synthesis strategy as an essential yet previously underexplored parameter for achieving high-performance ORR electrocatalysts. This study sheds light on the design of low-cost, high-performance catalysts for DMFCs.

2 Experimental

2.1 Chemicals and materials

Cobalt(II) nitrate hexahydrate ($\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$, >99%, Innochem), manganese(II) nitrate tetrahydrate ($\text{Mn}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$, >97.0%, Sigma Aldrich), iron(III) nitrate nonahydrate ($\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, >98.0%, Alfa Aesar), perfluorosulfonic acidpolytetrafluoroethylene copolymer (Nafion, Alfa Aesar), potassium hydroxide (KOH, >99%, Aladdin), ammonium fluoride (NH_4F , >98%, Aladdin), melamine (>99%, Aladdin), commercial platinum (20 wt%) loaded on carbon black (Alfa Aesar), hydroxylated multi-walled carbon nanotubes (CNT, XFNano), N,N-dimethylformamide (DMF, >99.0%, Kanto Chemical), anhydrous ethanol (>99.8%, Aladdin), and aqueous ammonia (25-30%, Innochem) were used as received without further purification.

2.2 Preparation of the catalysts and precursors

Synthesis of MnFeCo-LDH@CNT: CNT (30 mg) was dispersed in 50 mL of DMF under ultrasonication. Subsequently, 0.4 mmol $\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$, 0.2 mmol $\text{Mn}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$, 0.2 mmol $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ and 1.8 mmol NH_4F were added to the suspension. The mixture was stirred at 800 rpm for 1 h until a homogeneous solution was obtained. Then, 18 mL ammonia solution was slowly added dropwise to the suspension, followed by continuous stirring at room temperature for 4 h. The mixture was then allowed to stand for 10 h to complete the precipitation process. The precipitate was collected, washed three times with ethanol water mixtures (1:1), and vacuum-dried (DZF-6020) to obtain MnFeCo LDH loaded on CNT (MnFeCo-LDH@CNT).

Synthesis of MFC@N-CNT: The as-prepared MnFeCo-LDH@CNT was dispersed in 15 mL ethanol under ultrasonication, followed by the addition of 0.05 g melamine. The mixture was stirred at room temperature for 1 h and vacuum-dried at 60 °C to remove residual ethanol and water. The dried sample was thoroughly ground and then heated in a tubular furnace (OTF-1200X) at 800 °C for 2 h. The obtained product was washed three times with ethanol water mixtures (1:1) and vacuum-dried at 60 °C to obtain Mn-doped $\text{Fe}_3\text{O}_4/\text{Co}$ loaded on N-doped CNT (MFC@N-CNT).

Synthesis of MFC@CNT: The MnFeCo-LDH@CNT precursor was thoroughly ground and heated at 800 °C for 2 h without the addition of melamine. The obtained product was washed three times with ethanol water mixtures (1:1) and vacuum-dried at 60 °C to obtain Mn-doped $\text{Fe}_3\text{O}_4/\text{Co}$ loaded on CNT without N doping (MFC@CNT). This control sample was prepared to reveal the effect of N doping on the microstructure and electrocatalytic performance.

Synthesis of MFC@N-CNT with different metal loadings: To investigate the influence of metal content, a series of samples were synthesized following the above procedure, using double or half concentrations of metal salts, respectively. The obtained products were named MFC(*2)@N-CNT and MFC(*0.5)@N-CNT. These samples were used to evaluate the effects of metal loading on electrocatalytic performance.

Synthesis of MC@CNT: MnCo LDH@CNT was prepared following the same procedure as MnFeCo-LDH@CNT without adding $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, followed by calcination and vacuum drying without melamine (same as MFC@CNT). This control sample was used to compare the electrocatalytic performance between bimetallic and trimetallic catalysts loaded on CNT.

2.3 Sample characterization

The crystal phases of the catalysts were characterized by powder X-ray diffraction (XRD) using a Philips X'pert Pro MPD diffractometer with $\text{Cu K}\alpha$ radiation. The morphologies and structures of the catalysts were examined by scanning transmission electron microscopy (STEM) on an FEI Talos F200S microscope operated at 200 kV, equipped with a high-angle annular dark-field (HAADF) detector and energy-dispersive X-ray spectroscopy (EDX), and by scanning electron microscopy (SEM) on a Hitachi S8010 at 10 kV. X-ray photoelectron spectroscopy (XPS)

was conducted on a Shimadzu Axis Ultra spectrometer with Mg K α radiation. The elemental compositions were determined by inductively coupled plasma atomic emission spectroscopy (ICP-AES, Jarrel-Ash ICAP-9000).

2.4 Electrochemical measurements

The electrochemical performance of the catalysts was evaluated using a CHI 760E electrochemical workstation with a three-electrode system. The system consisted of a catalyst-coated glassy carbon rotating disk electrode (RRDE-3A) as the working electrode (WE), a Pt wire as the counter electrode (CE) and a saturated calomel electrode as the reference electrode (RE).

For electrode preparation, 2.0 mg catalyst was dispersed in 0.5 mL ethanol water mixtures (1:1) with 7.5 μ L Nafion binder under ultrasonication to form a homogeneous slurry. The slurry was drop-casted onto the WE and dried naturally.

Cyclic voltammetry (CV) and linear sweep voltammetry (LSV) were performed under N₂ or O₂ atmospheres. The electrolyte was saturated by bubbling the corresponding gas for 30 min before measurements. CV was conducted at 50 mV s⁻¹ and LSV at 20 mV s⁻¹.

3 Results and discussion

To investigate the crystal structure of the prepared catalysts, the XRD patterns of MFC@N-CNT and MC@CNT were performed. As shown in Fig. 1A, the XRD pattern of MFC@N-CNT exhibits a characteristic diffraction peak at 25.9°, which can be assigned to the (002) peak of CNT. Distinct diffraction peaks located at 32.5°, 35.4°, 38.1° and 64.7° correspond to the (103), (311), (004) and (314) peaks of Fe₃O₄, respectively. Diffraction peaks corresponding to metallic Fe or other iron oxide compounds are not observed, indicating that Fe species are predominantly converted into Fe₃O₄ during the synthesis process. In addition, diffraction peaks at 44.2° and 51.5° are attributed to the (111) and (200) planes of metallic Co. The poor consistency of the detected peaks with the characteristic peaks of cobalt oxide further suggests that cobalt mainly exists in the metallic state. Notably, the diffraction peaks of Mn₃O₄ strongly overlap with those of Fe₃O₄ and share identical lattice planes, making it difficult to distinguish Mn₃O₄ and Fe₃O₄ solely by XRD. Considering the relatively low Mn content determined by subsequent inductively coupled plasma (ICP) analysis, it is reasonable to propose that Mn is incorporated into the Fe₃O₄ lattice, forming a Mn-doped Fe₃O₄ intermetallic compound. The contents of Mn and Fe species are further investigated by XPS, ICP and EDX mapping analyses (vide infra).

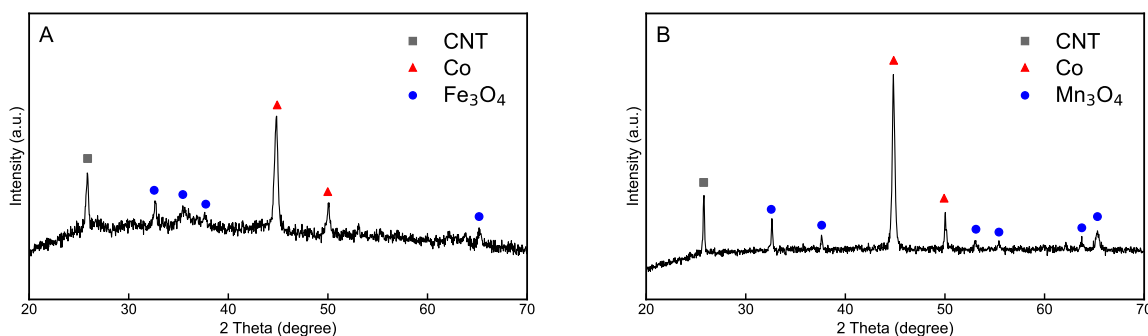


Figure 1: XRD patterns of (A) MFC@N-CNT and (B) MC@CNT.

The XRD pattern of MC@CNT (Fig. 1B) displays a similar CNT (002) reflection at 25.9°, along with characteristic peaks of metallic Co at 44.2° and 51.5°, corresponding to the (111) and (200) planes. Additional reflections at 32.5°, 38.1°, 51.6°, 54.8°, 63.2° and 64.7° are attributed to the (103), (004), (422), (511), (116) and (314) peaks of Mn₃O₄, respectively. The

absence of peaks corresponding to metallic Mn or other manganese oxides indicates that most Mn exists as Mn_3O_4 . These results confirm that Co and Mn exist as metallic Co and Mn_3O_4 , respectively, in MC@CNT. Excluding the peaks derived from Co and CNT, obvious differences in the diffraction peaks between MFC@N-CNT and MC@CNT are observed, suggesting distinct crystal orientations of the Mn-doped Fe_3O_4 intermetallic compound and Mn_3O_4 . The structural differences are considered to influence the electrocatalytic performance of the two catalysts.

Figure 2 presents the XPS survey and high-resolution spectra of MFC@N-CNT. The survey spectrum (Fig. 2A) confirms the presence of C, N, O, Co, Mn, and Fe elements. In the Co 2p spectrum (Fig. 2B), two prominent peaks at 780.0 and 795.8 eV, corresponding to Co 2p_{3/2} and Co 2p_{1/2}, along with their satellite peaks, indicate the presence of Co(II)[22]. This phenomenon can be explained by the surface oxidation of metallic Co nanoparticles upon exposure to air, since the XRD results suggest that Co predominantly exists in the metallic state. The Fe 2p spectrum (Fig. 2C) shows characteristic peaks at 710.6 and 724.1 eV, which are attributed to Fe 2p_{3/2} and Fe 2p_{1/2} in Fe_3O_4 , consistent with the XRD analysis[23]. The peaks of Mn 2p_{3/2} and Mn 2p_{1/2} are observed at 642.9 and 653.7 eV, confirming the presence of Mn_3O_4 (Fig. 2D)[24].

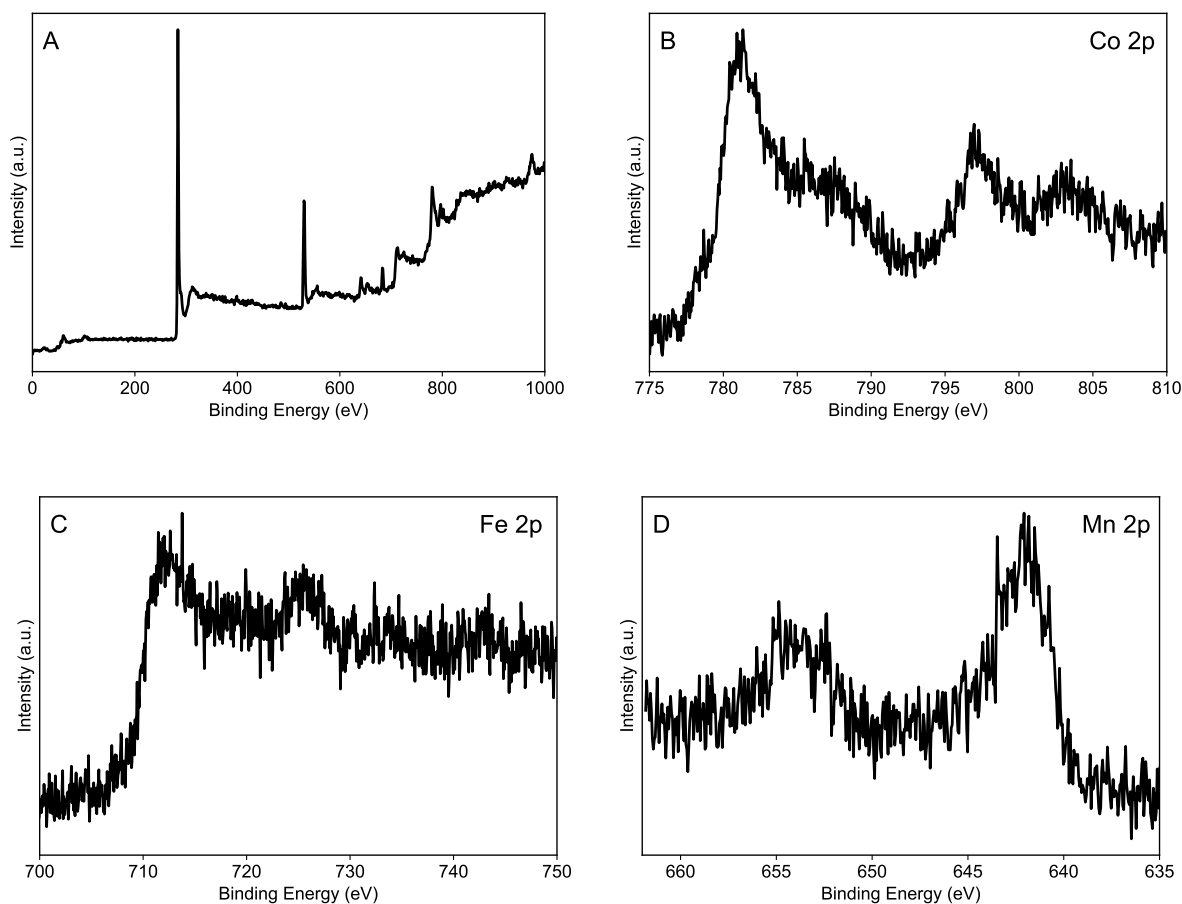


Figure 2: (A) XPS survey spectrum and high-resolution (B) Co 2p, (C) Fe 2p, (D) Mn 2p spectra of MFC@N-CNT.

Quantitative analysis by ICP reveals that the elemental contents of metallic Co, Fe_3O_4 , and Mn_3O_4 are 17.6%, 18.7%, and 12.7%, respectively. Considering the XPS results and the structural similarity between Fe_3O_4 and Mn_3O_4 , it is reasonable to propose that Mn is doped into the Fe_3O_4 lattice as a dopant, rather than forming a separate phase.

Figure 3 shows the SEM images of the MnFeCo-LDH@CNT precursor, MFC@N-CNT, and MFC@CNT. As shown in Fig. 3A and 3B, the precursor exhibits a composite structure of

tube-like CNT and flake-like LDH nanoflakes, where CNT are located on the surface of LDH. Such an interconnected morphology is expected to facilitate the embedding of metal/metal oxide nanoparticles into the CNT framework after calcination and suppress the detachment of catalytic particles, leading to enhanced structural stability of MFC@N-CNT. At this magnification, the layered structure and wrinkles of LDH cannot be clearly observed, which will be further investigated by STEM (*vide infra*).

After calcination (Fig. 3C and 3D), the LDH precursor is converted into smaller, well-dispersed metal/oxide nanoparticles, which are uniformly distributed and partially embedded in the CNT framework. In contrast, the SEM images of MFC@CNT without N doping (Fig. 3E and 3F) show that the catalytic particles exhibit a stronger tendency to aggregate. This difference can be attributed to the addition of melamine, which increases the nitrogen content of CNT during the calcination process and effectively suppresses nanoparticle agglomeration during formation. Moreover, previous studies have suggested that nitrogen incorporation may also enhance ORR electrocatalytic activity, which will be evaluated in subsequent electrochemical measurements[25].

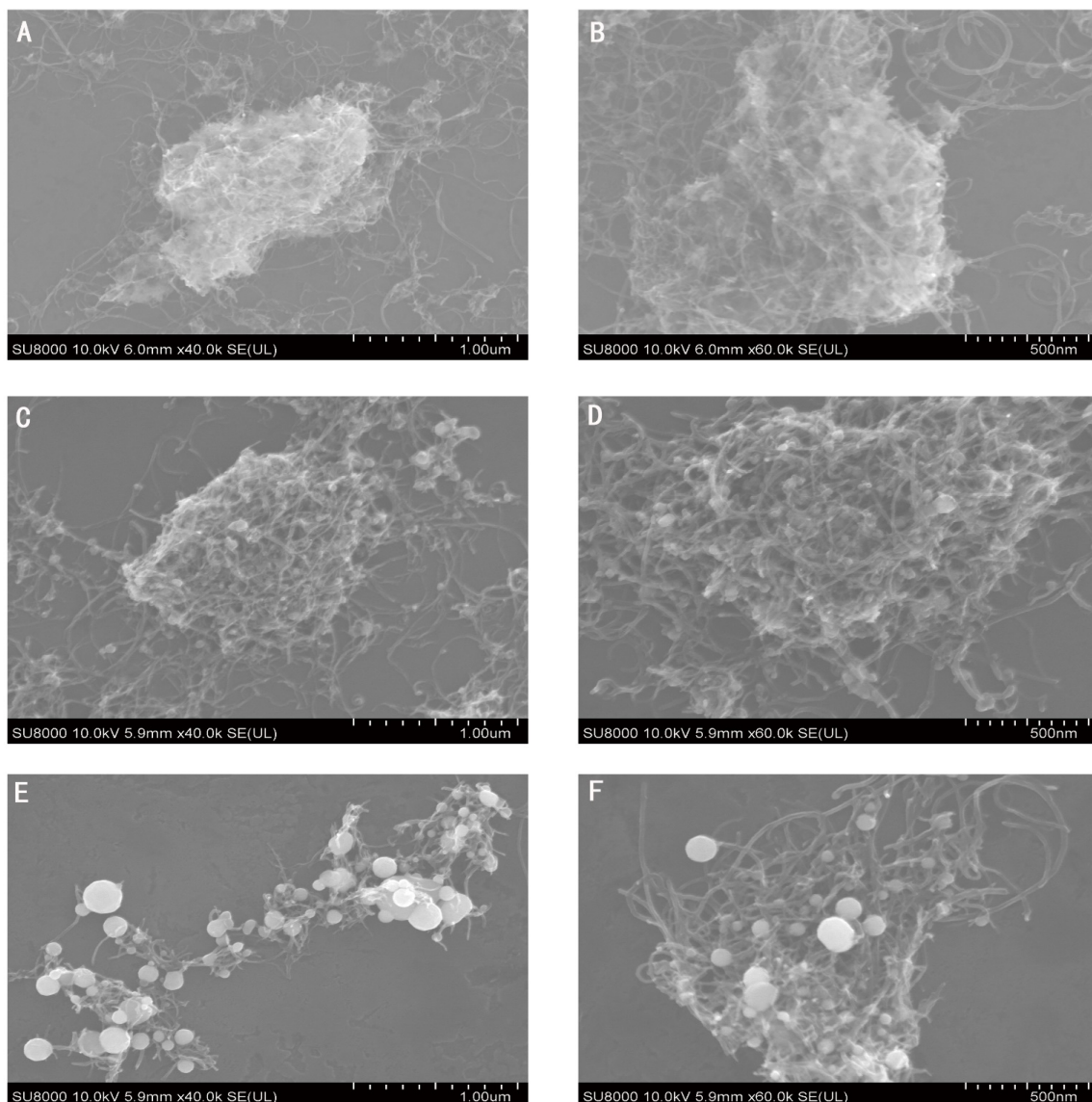


Figure 3: SEM images of (A)(B) MnFeCo-LDH@CNT, (C)(D) MFC@N-CNT and (E)(F) MFC@CNT.

The STEM image of the MnFeCo-LDH@CNT precursor (Fig. 4) shows the layered and wrinkled morphology of LDH nanoflakes. In contrast, the STEM image of MFC@N-CNT (Fig. 5A) shows that the catalytic nanoparticles are uniformly and densely distributed in the CNT framework, although some aggregation is observed. The size of the nanoparticles is estimated to range from about 20 to 100 nm. High-resolution TEM (HRTEM, Fig. 5B) further shows the well-defined lattice fringes on the surface of the MFC@N-CNT catalyst. Lattice spacings of 0.249 and 0.291 nm are observed, corresponding to the (311) and (220) planes of Fe_3O_4 , respectively. These results provide direct evidence for the existence of Fe_3O_4 in the CNT framework.

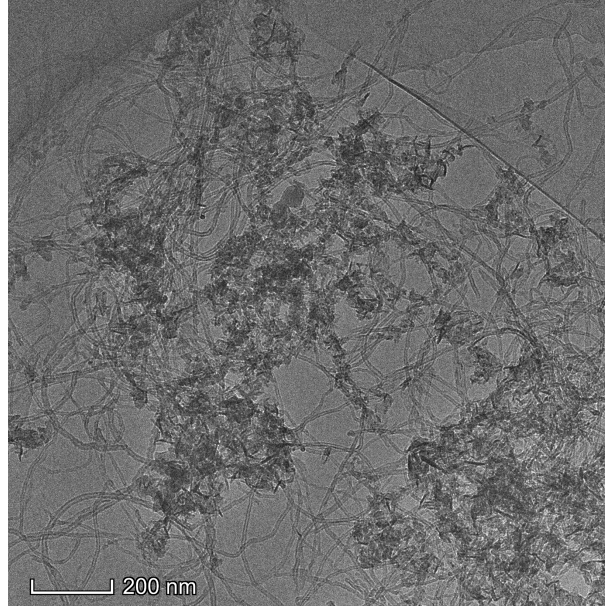


Figure 4: STEM image of MnFeCo-LDH@CNT.

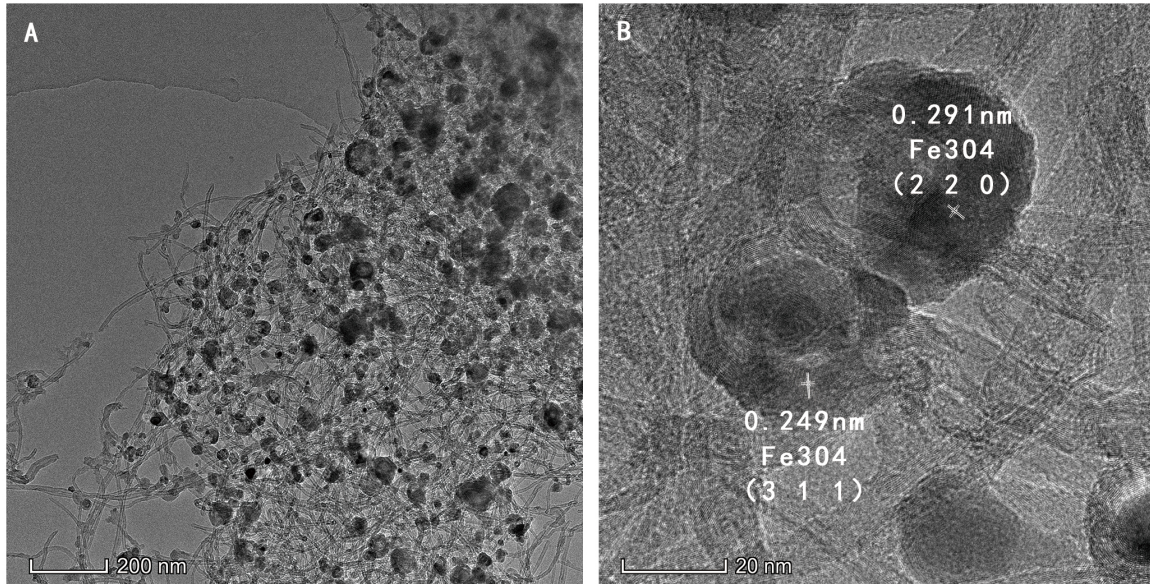


Figure 5: (A)STEM and (B) HRTEM images of MFC@N-CNT.

Figure 6 shows the EDX elemental mapping images of MFC@N-CNT, indicating the spatial distribution of elements within the catalyst. Fe and Co are homogeneously dispersed throughout the CNT framework, consistent with the distribution of catalyst nanoparticles (bright regions in Fig. 6A). The consistent spatial distribution of Fe and Co suggests good integration of Fe_3O_4 and

Co within the nanoparticles. The mapping image of Mn confirms its successful incorporation into the material; however, the relatively dispersed distribution of Mn signals, combined with the indistinguishable XRD peaks from Fe_3O_4 , further implies that Mn is incorporated into the Fe_3O_4 lattice via substitution of Fe atoms, rather than forming a separate Mn_3O_4 phase.

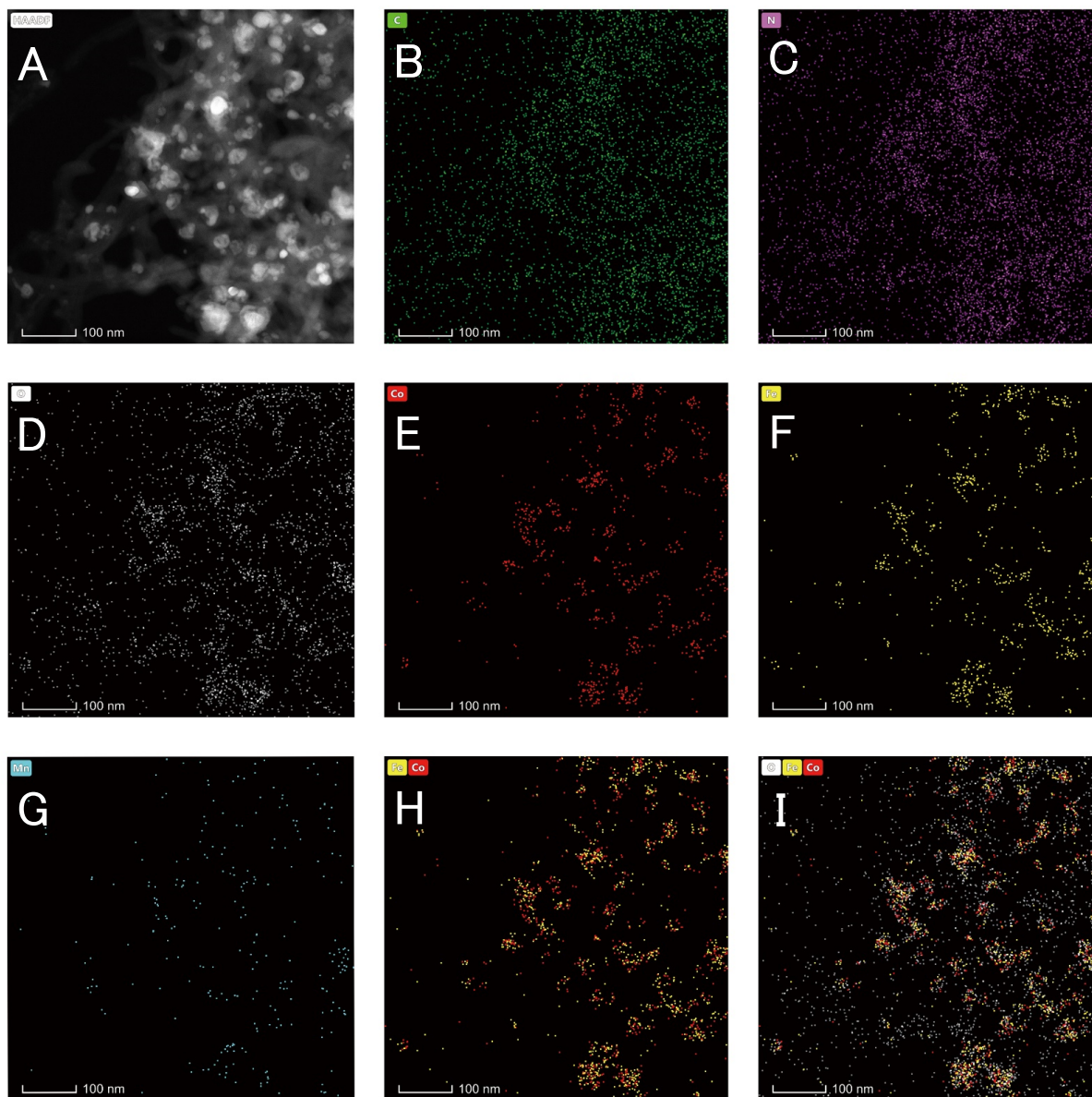


Figure 6: (A) STEM image of MFC@N-CNT. The EDX mapping of (B) C, (C) N, (D) O, (E) Co, (F) Fe, (G) Mn, (H) Fe, Co and (I) O, Co, Fe elements.

Figure 7 shows the CV curves of MFC@N-CNT and commercial Pt/C catalyst-coated electrodes obtained under N_2 - and O_2 -saturated electrolytes. In the N_2 -saturated electrolyte, both MFC@N-CNT and Pt/C electrodes exhibit smooth CV curves within the potential range of 0.61.0 V, while no ORR peaks are observed. In contrast, under O_2 -saturated conditions, obvious cathodic peaks corresponding to ORR are observed at similar potentials in both catalytic systems. These results confirm that MFC@N-CNT exhibits ORR electrocatalytic activity comparable to that of the traditional Pt/C electrode.

To elucidate the synergistic effect of Fe and Mn species, the electrocatalytic performances of MFC@CNT and MC@CNT were compared. To exclude the influence of nitrogen doping, pristine CNT was used as the carbon substrate. As shown in Fig. 8A, both catalysts exhibit similar

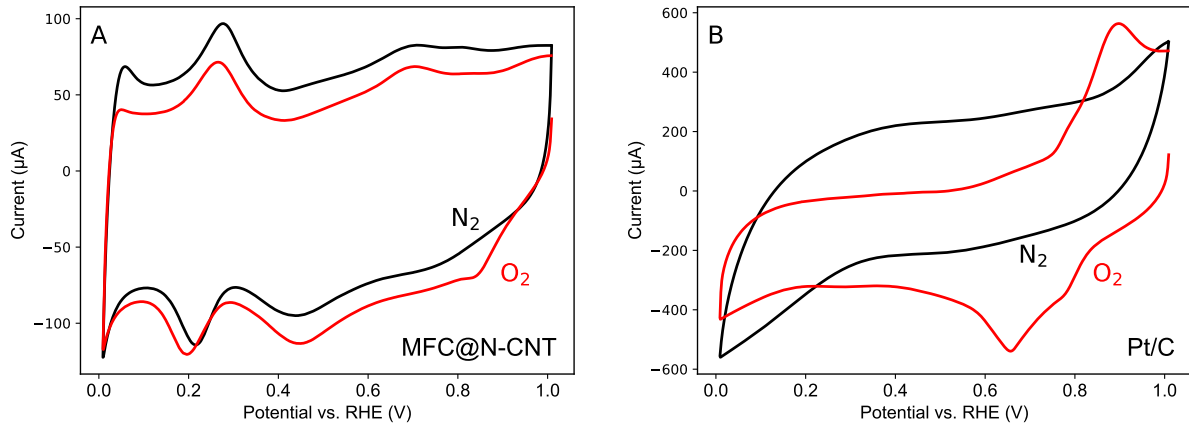


Figure 7: CV recorded on (A) MFC@N-CNT and (B) Pt/C working electrodes under (black) N_2 - and (red) O_2 -saturated conditions. The scan rate was 50 mV s^{-1} .

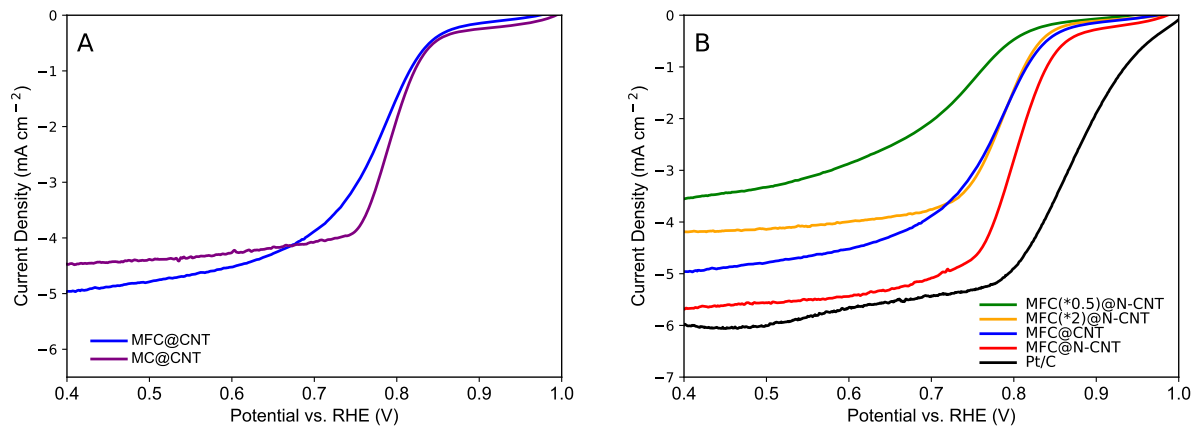


Figure 8: LSV recorded on (A) (blue) MFC@CNT, (purple) MC@CNT and (B) (black) Pt/C, (red) MFC@N-CNT, (blue) MFC@CNT, (orange) MFC(*2)@N-CNT, (green) MFC(*0.5)@CNT working electrodes under O_2 -saturated conditions. The scan rate was 20 mV s^{-1} and the rotating rate was 1600 rpm.

ORR onset potentials, while the $E_{1/2}$ values of MFC@CNT and MC@CNT are 0.771 V and 0.765 V, respectively. The higher $E_{1/2}$ of MFC@CNT indicates that the incorporation of Fe_3O_4 , Mn_3O_4 and Co enhances ORR electrocatalytic activity. This improvement may originate from differences in the preferred crystal growth planes between the Mn-doped Fe_3O_4 intermetallic compound and Mn_3O_4 . Subsequent studies will investigate the $Fe_3O_4/Co@CNT$ system to clarify whether Mn incorporation modifies the dominant crystal growth planes of Fe_3O_4 and enhances the electrocatalytic activity of MFC@N-CNT.

Fig. 8B compares the LSV curves of MFC@N-CNT with MFC(*2)@N-CNT, MFC(*0.5)@N-CNT, MFC@CNT, and Pt/C under O_2 -saturated conditions. Although all samples show comparable onset potentials for ORR, their half-wave potentials ($E_{1/2}$) are distinguished as 0.793 V, 0.771 V, 0.784 V, 0.720 V, and 0.840 V, respectively. Notably, MFC@N-CNT exhibits a more positive $E_{1/2}$ than MFC@CNT and the control samples with doubled or halved metal contents, indicating that MFC@N-CNT shows optimized ORR electrocatalytic activity. However, its performance remains slightly inferior to that of commercial Pt/C, suggesting that further optimization is necessary.

Figure 9 illustrates the cycling stability of MFC@N-CNT (Fig. 9A), MFC@CNT (Fig. 9B)

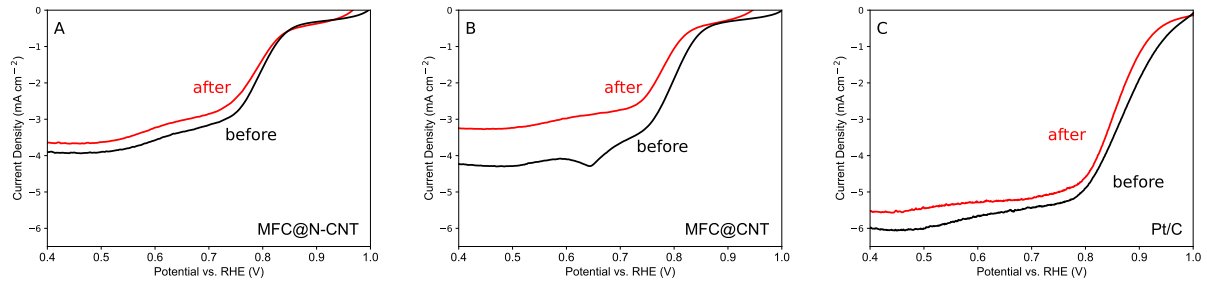


Figure 9: LSV recorded on (A) MFC@N-CNT, (B) MFC@CNT and (C) Pt/C working electrodes under O_2 -saturated conditions (black) before and (red) after 500 cycles CV measurements. The scan rate was 20 mV s^{-1} and the rotating rate was 1600 rpm.

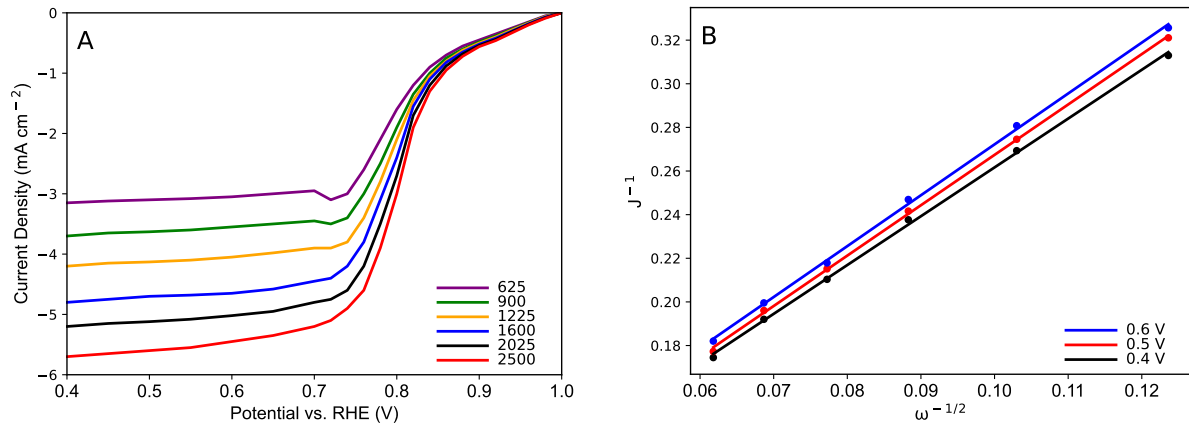


Figure 10: (A) LSV recorded on MFC@N-CNT working electrode under O_2 -saturated conditions at different rotating rates. The scan rate was 20 mV s^{-1} . (B) K-L plots obtained at potentials of (black) 0.4 V, (red) 0.5 V and (blue) 0.6 V.

and Pt/C (Fig. 9C) catalysts. ORR degradation was evaluated by comparing LSV curves before and after 500 CV cycles. The corresponding decreases in $E_{1/2}$ are 8, 14, and 11 mV for MFC@N-CNT, MFC@CNT, and Pt/C, respectively, indicating that MFC@N-CNT exhibits higher cycling stability. The enhanced stability is likely attributed to the smaller nanoparticle size and stronger attachment to the CNT network, which suppress nanoparticle detachment. Consequently, MFC@N-CNT is expected to maintain ORR activity over prolonged operation, thereby extending the lifetime of DMFCs.

To reveal the ORR pathway of the MFC@N-CNT catalyst, the electron transfer number is calculated using the Koutecky-Levich (K-L) plot[26]. The current density of ORR can be described by the K-L equation:

$$\frac{1}{J} = \frac{1}{J_k} + \frac{1}{J_d} + \frac{1}{J_f}$$

Considering that $\frac{1}{J_f}$ is negligible, the equation can be simplified to:

$$\frac{1}{J} = \frac{1}{J_k} + \frac{1}{J_d}$$

For a given reaction, J_d can be treated as a constant and expressed as:

$$J_d = 0.620 n F A C_0 D^{2/3} \omega^{1/2} \nu^{-1/6}$$

By defining $0.620 n F A C_0 D^{2/3} \omega^{1/2} \nu^{-1/6}$ as K , the equation can be simplified to:

$$\frac{1}{J} = \frac{1}{J_k} + \frac{1}{K \omega^{1/2}}$$

Thus, plotting $\frac{1}{J}$ versus $\omega^{-1/2}$ yields a straight line with a slope of $\frac{1}{K}$, from which the electron transfer number (n) can be calculated[27]. Here, J is the measured current density, J_k

is the kinetic current density, J_d is the limiting current density, and J_f is the diffusion-limited current density; n is the electron transfer number of ORR; F is the Faraday constant; A is the electrode surface area; C_0 is the saturated O_2 concentration; D is the diffusion coefficient of O_2 ; ω is the rotation rate; and ν is the dynamic viscosity of the electrolyte[28].

As shown in Fig. 10B, K-L plots obtained at potentials of 0.4 V, 0.5 V and 0.6 V yield electron transfer numbers of 4.06, 3.94, and 3.90, respectively. These values are close to 4, corresponding to the ideal four-electrons ORR pathway, implying minimal side reactions and optimized energy conversion efficiency of the MFC@N-CNT electrocatalyst.

4 Conclusion

In this work, a rationally designed MFC@N-CNT material was developed as a cost-effective and efficient electrocatalyst for ORR. Through the integration of LDHs precursors, controlled thermal transformation, and in situ nitrogen doping, a well-dispersed and high-performance nanocatalyst was successfully constructed. Structural characterization confirms the formation of Mn-incorporated Fe_3O_4 coupled with metallic Co nanoparticles uniformly embedded within an N-doped CNT network, providing abundant active sites and facilitating rapid charge transport.

Electrochemical measurements demonstrate that MFC@N-CNT exhibits competitive ORR activity compared with commercial Pt/C and superior long-term cycling stability. The four-electrons transfer pathway further highlights its high selectivity and efficiency. Control experiments reveal that both multimetal synergy and nitrogen doping play important roles in optimizing catalytic performance and cycling stability.

Based on the MFC@N-CNT system, this study highlights the importance of synthesis strategy in tailoring catalyst structure and electrochemical properties. The proposed approach provides a versatile platform for designing low-cost, high-performance electrocatalysts. The development of such earth-abundant and stable ORR catalysts represents an important step toward the commercialization of fuel cell technologies, especially DMFCs.

AI Process Disclosure

This research was conducted with AI assistance. The complete process log, including all AI agent transcripts, prompts, and a timestamped record of human decisions, is available in the [process-log directory](#) of the paper repository.

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